Team Number.	apmcm2300899
Problem Chosen.	C

Advancing NEVs: A Comprehensive Analysis and Quantitative Modeling

Abstracts:

The ongoing global electrification transformation in the automotive industry is propelling the new energy vehicle sector towards a comprehensive stage of market development. Stakeholders advocate enhanced international collaboration in new energy vehicles to foster mutual benefits [1]. Since 2011, the Chinese government has actively promoted new energy electric vehicles through preferential policies. This paper establishes quantitative models to assess the current status and future trends, aiming to comprehensively understand and address challenges in the sector.

Problem 1, Utilizing the **entropy weight method** [2], this study quantifies the weight and contribution of each factor influencing the development of new energy electric vehicles. **Multiple linear regression** [3] is employed to establish the specific impact of government policies, technological innovation, economic factors, charging infrastructure, and competitors' behavior on sales volume, validated through the **least squares method**.

Problem 2, A **grey prediction model** [4], based on historical data incorporating sales, government support, and technological innovation, predicts an annual increase in new energy vehicle sales over the next decade.

Problem 3, A **multivariate nonlinear regression model** [5] is fitted using a **time-series dataset**, incorporating global sales and market share of new energy and traditional vehicles, oil prices, charging infrastructure, etc. Results indicate that the development of new energy vehicles inhibits traditional vehicle industry growth, driving a cleaner and more efficient direction for the entire automobile industry.

Problem 4, The **Kruskal-Wallis H-test** [6] assesses the impact of boycott policies in other countries on China's new energy vehicles, considering boycott behavior, growth rate, and policy effects on demand. A **supply and demand model** [7], estimated through the **least squares method**, predicts limited market access, price competitiveness, image damage, corporate strategy adjustment, and industrial restructuring as outcomes.

Problem 5, Investigating the environmental impacts throughout the production, use, and disposal phases, a **system dynamics model** [8] combined with **life cycle analysis (LCA)** [9] and the **Runge-Kutta method** [10] reveals that electric vehicles improve air quality by reducing tailpipe emissions and contribute to energy consumption restructuring.

Problem 6, Based on the above findings, an open letter to the public highlights the positive impact of new energy vehicles and emphasizes the significant contribution of the electric vehicle industry at the national level.

Keywords: entropy weighting method, grey prediction model, multivariate nonlinear regression, Kruskal-Wallis H-test, system dynamics model

Contents

Abstracts: 1
I. Introduction 2
1.1 Background 2
1.2 Problem Restatement 2
II. Analysis of the problem 3
2.1 Question one 3
2.2 Question two 3
2.3 Question three 3
2.4 Question four 4
2.5 Question five 4
2.6 Question six 4
III. Model assumptions 4
IV. Symbol setting 5
V. Modelling and solving 5
5.1 Problem 1 modelling and solution 5
5.1.1 Entropy weight method 5
5.1.2 Multiple linear regression 6
5.1.3 Conclusion 8
5.2 Problem 2 modelling and solution 9
5.2.1 Data collection and processing 9
5.2.2 Establishment of a grey prediction model 9
5.2.3 Model solving process 9
5.2.4 Conclusion 10
5.3 Problem 3 modelling and solution 11
5.3.1 Data collection and analysis 11
5.3.2 Modelling multivariate non-linear regression 12
5.3.3 Conclusion 13
5.4 Problem 4 modelling and solution 13
5.4.1 Modelling requirements 13
5.4.2 Supply and demand modelling solutions 14
5.4.3 Decision analysis and game theory modelling 15
5.4.4 Conclusion 16
5.5 Problem 5 modelling and solution 17
5.5.1 Modelling system dynamics 17
5.5.3 Solving with Life Cycle Assessment (LCA) 18
5.5.3 Conclusion 20
5.6 Question 6 - Open Letter 21
VI. Model evaluation 22
6.1 Model benefits 22
6.2 Modelling shortcomings 23
VII. Model roll-out 23
VIII. References 23

I. Introduction

1.1 Background

With the evolving societal landscape and economic structural transformations, energy-related issues and environmental conservation have garnered heightened attention. Global focus on energy transition and carbon neutrality, vital strategies to combat climate change, has surged [11]. Grounded in the paradigm of upgrading conventional fossil fuels and optimizing the utilization of new energy sources, new energy vehicle technologies are rapidly advancing. In the context of China, a major player in the global automotive market, the development of new energy vehicles assumes significance not only for environmental and energy security concerns but also as a pivotal component of national strategy and industrial enhancement. Recent years have witnessed the issuance of comprehensive policies and plans by the Chinese government, encompassing macro-level planning, manufacturing standards, infrastructure development, purchase subsidies, supportive tariffs, and more. The trajectory of new energy vehicle development in China is shaped by multifaceted factors, encompassing technological advancements, cost efficiencies, consumer preferences, policy backing, and the global economic milieu.

Furthermore, the imperative to address the escalating discord between fuel supply and demand, coupled with environmental degradation, has prompted major automobile-producing nations worldwide to position new energy vehicles as a key component of their low-carbon transport development strategy. Accelerating the research and development and industrialization of new energy vehicles emerges as a critical pathway for nations to address energy security, environmental preservation, and the transformation and upgrading of the automotive industry. Consequently, a comprehensive understanding of the diverse factors influencing new energy vehicle development assumes paramount importance in formulating judicious policies and steering industrial progress.

1.2 Problem Restatement

Problem 1: Factor Analysis for New Energy Electric Vehicle Development in China
Utilize the entropy weighting method to collect and quantify data.
Apply multiple linear regression to ascertain the impact of each factor.
Conduct an error analysis through the least squares method.
Problem 2: Forecasting China's New Energy Electric Vehicle Industry
Aggregate data encompassing sales, policy support, and technological innovation.
Construct a grey prediction model for a decade-long projection of China's new energy electric vehicle development.
Problem 3: Global Impact Assessment of New Energy Electric Vehicles
Acquire global data on sales, market share, oil prices, and charging infrastructure.
Develop a multivariate non-linear regression model to evaluate the influence of new

energy electric vehicles on the global traditional energy vehicle industry.

Problem 4: Boycott Impact on China's New Energy EV Market

Collect data on boycott behaviors and growth rates from other countries.

Perform a Kruskal-Wallis H-test and build a supply and demand model to quantify the boycott's impact.

Estimate changes in China's new energy EV market post-boycott using the least squares method.

Problem 5: Urban Electrification Impact Modeling in Xiamen

Assume Xiamen as a city of one million people and collect data on urban electrification impact.

Develop a system dynamics model based on vehicles, traffic flow, pollutant emissions, air quality, noise levels, and energy mix.

Utilize the Runge-Kutta method to solve and assess impact results through life cycle assessment.

Problem 6: Public Communication Based on Urban Electrification Findings

Draft an open letter to the public, highlighting positive impacts of new energy vehicles. Emphasize the substantial contributions of the electric vehicle industry on a national scale, informed by conclusions from above.

II. Analysis of the problem

2.1 Question one2.2 Question two

(1) Objective: This inquiry examines the principal determinants influencing the advancement of new energy electric vehicles in China, encompassing government policy, technological innovation, economic factors, charging infrastructure, consumer perceptions, and competitor behavior.

(2) Methodology: Employ the entropy weighting method to establish factor weights. Calculate the information entropy and assess uncertainty for each factor.

Conduct multiple linear regression, utilizing new energy electric vehicle sales as the dependent variable and government policy, technological innovation, economic factors, charging infrastructure, consumer perception, and competitors' behavior as independent variables.

Model coefficients provide specific influences on sales volume, with positive, negative, and magnitude indicating direction and relative strength. Validate results using the least squares method.

(3) Conclusion: The primary determinants impacting new energy electric vehicle development in China comprise government policies, technological innovation, economic factors, charging infrastructure, and competitors' behavior.

2.3 Question three

(1) Data Collection: Conducted extensive data gathering, including key global information on sales volume and market share for both new and conventional energy

vehicles, oil prices, and the number of charging piles, forming a comprehensive time series dataset.

(2) Model Development: Established a multivariate non-linear regression model utilizing historical data on new energy vehicle sales and traditional energy sales volume. Predicted the sales volume of new energy electric vehicles for the next decade.

(3) Impact Analysis: Analyzed and interpreted model results to assess the impact of global new energy electric vehicle sales on conventional vehicle sales volumes.

2.4 Question four

(1) Boycott Policy Impact Analysis: Conducted a Kruskal-Wallis H-test to assess the impact of boycott policies in other countries on new energy vehicles in China, considering boycott behavior and growth rate.

(2) Modeling Demand and Supply: Modeled demand and supply, factoring in the influence of price, publicity, and technology. Estimated model parameters through the least squares method to quantify policy impact on demand and supply.

(3) Impacts on China's New Energy Vehicles: Identified impacts of boycott policies, including limited market access, price competitiveness, image damage, enterprise strategy adjustment, and industrial structure adjustment.

2.5Question five

(1) City Dynamics Modeling: Assumed Xiamen, China, as a city of one million people. Modeled system dynamics considering interactions between variables such as vehicle numbers, traffic flow, emissions, air quality, noise levels, and energy mix. Solved a system of differential equations using the Runge-Kutta method, with model parameters calibrated and validated against actual data.

(2)Life Cycle Assessment (LCA): Conducted an LCA assessment encompassing emissions, resource use, manufacturing, use, and disposal phases. Calculated environmentally relevant indicators to evaluate ecological impacts of electric vehicles.

(3) Simulation Results: Concluded that electric vehicle promotion reduces tailpipe emissions, improves city air quality, alters energy consumption structure through traditional fuel substitution, and induces industrial chain restructuring through new energy electric vehicle recycling.

2.6 Question six

Based on insights from above, composed an open letter to the public emphasizing positive impacts of new energy vehicles and the significant contribution of the electric vehicle industry at the national level.

III. Model assumptions

1. Assuming that no major changes occur in the new energy vehicle industry and that the development trend remains unchanged.
2. It is assumed that the end-of-life volume of traditional fuel vehicles is converted into new energy vehicles according to a certain percentage [12].
3. Certain non-critical factors are not taken into account in the modelling process.

IV. Symbol setting

Symbol	Description	Unit
S	Sales of new energy electric vehicles.	units/year
P	Level of support from government policies.	index
T	Level of technological innovation.	index
E	Economic factors such as fuel prices.	RMB/liter
C	Degree of sophistication of charging infrastructure.	index
A	Level of consumer acceptance and awareness.	index
COP	Competitor behaviour.	index
$G_{propaganda}$	Policy Variables for Advocating Negative Impacts.	index
$G_{technology}$	Policy variables for technical barriers.	index
$M_{Gasoline}$	Manufacturing costs of conventional fuel vehicles.	RMB/vehicle
M_{NEVs}	Manufacturing costs of NEVs.	RMB/vehicle
$E_{Gasoline}$	Energy consumption for the use of conventional fuel vehicles.	liter/100 km
E_{NEVs}	Energy consumption for the use of NEVs.	kWh/100 km
$D_{Gasoline}$	Abandonment costs of conventional fuel vehicles.	RMB/vehicle
D_{NEVs}	Disposal costs of NEVs.	RMB/vehicle

V. Modelling and solving

5.1 Problem 1 modelling and solution

5.1.1 Entropy weight method

1) Model Definition

Entropy weighting method is a multi-indicator comprehensive evaluation method used to determine the weight of each indicator, based on the information entropy of each indicator. The following are the steps of using the entropy weight method to analyse and model the main factors in the development of new energy electric vehicles in China:

1. Identification of evaluation factors

Identify the main factors that influence the development of new energy electric vehicles in China. These include government policies, technological innovation, economic factors, charging infrastructure, consumer perceptions, and competitor behaviours, which we use as evaluation factors.

2. Data normalisation

The data for each factor was normalised to ensure that they were on the same scale.

This is because the entropy weighting method is sensitive to the scale of the data.

3. Calculate information entropy

The information entropy of each factor is calculated to represent the uncertainty of each factor. The information entropy is calculated by the formula:

$$E(X) = -\sum_{i=1}^{n} P_i \log_2(P_i)$$

Where P_i is the weight of the ith factor

4. Calculation of weights

The weights of each factor are calculated and the relationship between weights and information entropy is:

$$W_i = \frac{1 - E(X_i)}{k - \sum_{i=1}^{n} (X_i)}$$

where k is the number of normalised data and n is the number of factors

5. Developing mathematical models

Multiple linear regression model using the obtained weights and normalised factor values

2) Model solving

Determine the weight of each factor by entropy weighting method. By calculating the information entropy of each factor, we are able to understand the uncertainty of each factor, and then get the weight of each factor by the entropy weight method. In this way, we can quantify the degree of contribution of each factor to the development of new energy electric vehicles.

Normalised factor data:

$$\begin{bmatrix} 0.2 & 0.5 & 0.8 & 0.3 & 0.6 & 0.7 \\ 0.7 & 0.3 & 0.5 & 0.6 & 0.8 & 0.4 \\ 0.5 & 0.8 & 0.4 & 0.7 & 0.3 & 0.6 \\ 0.6 & 0.7 & 0.2 & 0.8 & 0.5 & 0.3 \\ 0.4 & 0.6 & 0.7 & 0.2 & 0.8 & 0.5 \\ 0.8 & 0.4 & 0.6 & 0.5 & 0.7 & 0.2 \end{bmatrix}$$

Rows one to six are government policy, technological innovation, economic factors, charging infrastructure, consumer perceptions, competitor behaviour

According to the solution results, it can be seen that: technological innovation, economic factors, and charging infrastructure occupy a higher proportion in the weighting, indicating that their contribution to the overall goal is relatively large. Government policy, consumer perception, and competitor behaviour occupy a lower proportion in the weights, but still have a certain contribution. Taken together, the weights of the factors are relatively balanced

5.1.2 Multiple linear regression

1) Model Definition

Multiple Linear Regression MLR [13] is a method commonly used in statistics and

machine learning to establish a relationship between a dependent variable (target variable) and multiple independent variables (features). In this method, we assume that the dependent variable is a linear combination of the independent variables and find this relationship by fitting the coefficients of the model.

2) Selection of influencing factors

This question uses multiple linear regression to establish the relationship between the sales volume of new energy electric vehicles and government policies, technological innovation, economic factors, charging infrastructure, consumer perceptions, and competitor behaviour. The coefficient (coefficient) of each factor explains the effect of that factor on the target variable (sales volume). The magnitude and sign of the coefficients help us understand the relative contribution of these factors to sales volume. The model makes predictions by learning the relationships in the training data.

3) Modelling

1. Multiple linear regression equations
2. Evaluation indicators:

$$S = a_0 + a_1 P + a_2 T + a_3 E + a_4 C + a_5 A + a_6 COP + \varepsilon$$

1. root mean square error RMSE:

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (S_i - \hat{S}_i)^2}{n}}$$

2. Coefficient of determination R^2 :

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (S_i - \hat{S}_i)^2}{\sum_{i=1}^{n} (S_i - \hat{S})^2}$$

3. Mean Absolute Error (MAE): MAE measures the average prediction error of the model.

$$MAE = \frac{\sum_{i=1}^{n} \left| S_i - \hat{S}_i \right|}{n}$$

4) Model fitting

1. Act I - Cross-validation

The performance of the model is evaluated on different subsets of data using K-fold cross-validation through machine learning scikit-learn models to check the generalisation ability of the model.

Conclusion:

The coefficients of the model indicate:

1. Similarity of influencing factors: the coefficient factors of all influencing factors in the model are relatively consistent in terms of their contribution to the development of new energy electric vehicles in China.

2. Smaller impact of competitors' behaviour: the coefficient of competitors' behaviour in the model is relatively small. That is, in the current model, the impact of competitor behaviour on the development of new energy electric vehicles is relatively small.

figure 1 Actual vs Predicted Sales

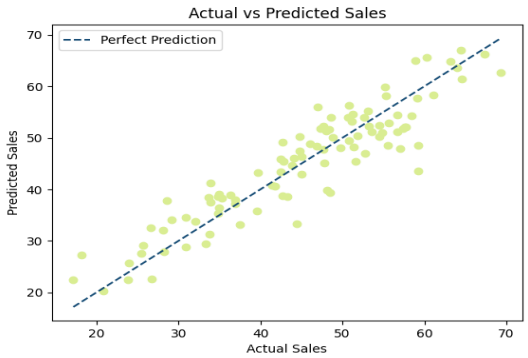

figure 2 Actual vs Predicted Sales

2.Method 2: Least Squares OLS

OLS is widely used to fit linear models to find the line of best fit by minimising the sum of squared residuals, which minimises the error between the predicted and actual values of the model. In this problem, the coefficients indicate the extent to which each factor contributes to the sales volume of new energy electric vehicles. The sign and magnitude of the coefficients provide information about the direction and relative importance of the influence on these factors.

Conclusion:

form 1 Multiple regression results

	OLS	Regression	Results	
Dep.	Variable.	y	R-squared.	0.974
Model.	OLS	Adj.	R-squared.	0.949
Method.	Least	Squares	F-statistic.	38.06
No.	Observations.	5	AIC.	23.38
Df	Residuals.	2	BIC.	22.21
Df	Model.	2		
Covariance	Type.	nonrobust		

5.1.3 Conclusion

Coefficients: Coefficients reveal consistent contributions of influencing factors, demonstrating no noticeable bias in their impact on the development of new energy electric vehicles in China.

Competitor Behavior: The coefficient for competitor behavior is relatively small, signifying a modest influence on the development of new energy electric vehicles.

Model Explanatory Power: With an R-squared of 0.974, Adj. R-squared of 0.949, and F-statistic of 38.06, the model effectively explains variables, exhibiting overall significance. Approximately 97.4% of the changes in the dependent variable are accounted for by the model.

AIC and BIC Analysis: AIC at 23.38 and BIC at 22.21 indicate a well-fitted model. Low values for both indicators suggest a favorable model fit, emphasizing its quality.

Main Influencing Factors: Government policies, technological innovation, economic factors, charging infrastructure, and competitor behavior emerge as the

primary factors influencing the development of new energy electric vehicles in China.

5.2 Problem 2 modelling and solution

5.2.1 Data collection and processing

We begin with extensive data collection, including but not limited to key information on new energy vehicle sales data, government policy support, and technological innovation. These data will be used as the basis of our modelling, forming a time series dataset.

For the time series data, we performed preliminary data processing, including trend term extraction and cumulative generation. This helped to make the data more suitable for application in grey forecasting models.

5.2.2 Establishment of a grey prediction model

We have chosen the grey prediction model [14] as a modelling tool. The model is based on a small amount of data and describes the development law of the system by establishing grey differential equations. The specific steps include:

1、Establishment of grey differential equations: According to the sales data of new energy vehicles, grey differential equations are established, which include grey derivatives, grey cumulants and so on.

2. Parameter estimation: The key parameters in the model are identified by parameter estimation of the grey differential equations.

3, model test: the establishment of grey prediction model test, to verify its effect on the fitting of historical data.

4. Forecasts for the next 10 years: Using the established grey forecasting model, we carry out forecasts of new energy electric vehicle sales for the next 10 years. This includes a quantitative analysis of trends and developments over the next few periods.

5. Analysis and Interpretation of Results: Finally, we provide in-depth analysis and interpretation of the model's forecast results to extract key development trends and factors. This will provide relevant stakeholders with important insights into the future of the new energy vehicle market and provide a scientific basis for decision-making.

5.2.3 Model solving process

Before building the grey prediction model GM, the time series is subjected to a rank-ratio test. If it passes the level-ratio test, it means that the series is suitable for constructing the grey model, and if it does not pass the level-ratio test, the series is "shifted and transformed", so that the new series meets the level-ratio test.

The grey prediction model should be tested to determine whether it is reasonable or not, only through the test of the model can be used for prediction, the system is mainly through the a posteriori difference ratio C value to test the grey prediction model.

1) Grade ratio test

form 2 Grade comparison test result table

Index item	original value	level ratio after translation	sequence value after translation	level ratio after translation
2012	1.3	-	2171.3	-
2013	1.8	0.722	2171.8	1
2014	7.5	0.24	2177.5	0.997
2015	32.9	0.228	2202.9	0.988
2016	50.2	0.655	2220.2	0.992
2017	76.8	0.654	2246.8	0.988
2018	124.7	0.616	2294.7	0.979
2019	120.6	1.034	2290.6	1.002
2020	132.3	0.912	2302.3	0.995
2021	350.7	0.377	2520.7	0.913
2022	687.2	0.51	2857.2	0.882
2023	723.1	0.95	2893.1	0.988

From the analysis of the above table, it can be obtained that all the rank ratio values of the level-transformed series are located in the interval (0.857, 1.166), indicating that the level-transformed series is suitable for constructing the grey prediction model.

2) Grey model construction

form 3 Grey model construction

Development coefficienta	grey action quantityb	posterior error ratio C value
-0.03	1946.201	0.22

From the analysis of the above table, it can be obtained that the a posteriori difference ratio value is 0.22 and the model is highly accurate.

5.2.4 Conclusion

1) Model fit prediction plot

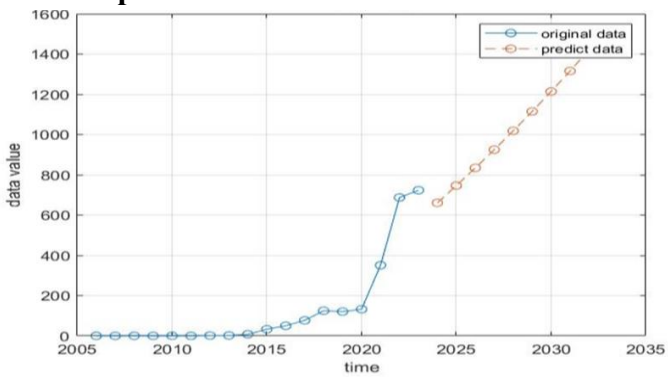

figure 3 Grey prediction trend chart

The above figure illustrates the fitted prediction plot of the grey prediction model.

2) Table of model prediction results

form 4 Model Prediction Results Table

prediction order	predictive value
1	660.827
2	746.280
3	834.313
4	925.004
5	1018.432
6	1114.680
7	1213.834
8	1315.981
9	1421.212
10	1529.619

The above table shows the table of prediction results of the grey prediction model.

According to the chart, the sales volume of new energy electric vehicles in China will increase every year, and the data for each year of the next ten years are predicted in the model forecast results table.

5.3 Problem 3 modelling and solution

5.3.1 Data collection and analysis

Collect the time series data of the sales volume of new energy electric vehicles and conventional vehicles in the past few years to ensure the accuracy and timeliness of the data. And make a time series graph of each variable.

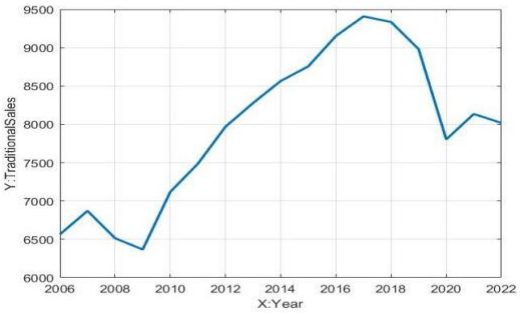

figure 4 Time series chart of global sales of traditional energy vehicles

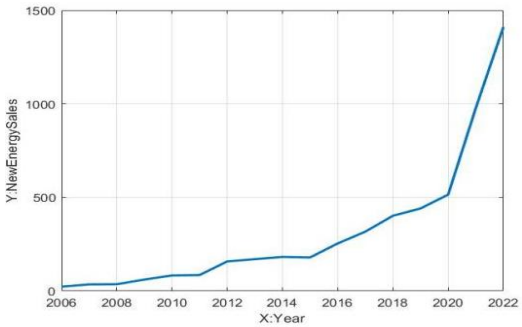

figure 5 Time series chart of global sales of new energy electric vehicles

Preliminary visualisation and analysis of the collected data was carried out by drawing bar charts to observe the relationship between the sales of new energy electric vehicles and the sales of conventional energy vehicles, and to make a preliminary judgement on the existence of a non-linear relationship.

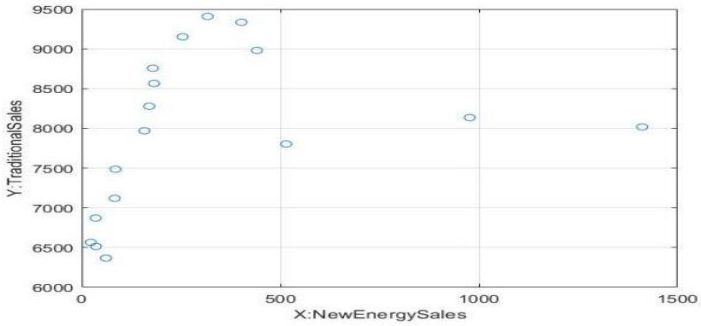

figure 6 The Sales of EVs and traditional energy vehicles

5.3.2 Modelling multivariate non-linear regression

1) Model assumptions

A multivariate polynomial relationship is assumed to exist between the sales of conventional energy vehicles, Y, and the sales of new energy electric vehicles, X.

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \ldots + \varepsilon$$

2) Model fitting

Use regression analysis tools to fit a quadratic polynomial regression model to the actual data by, for example, minimising the sum of squares of the residuals.

2) Model Evaluation

R-squared is a statistic used to measure the fit of a regression model. It represents the proportion of the variance in the dependent variable that is explained by the model.R-squared takes values between 0 and 1, with values closer to 1 indicating a better fit of the model to the data, and values closer to 0 indicating a worse fit of the model to the data.

$$R^2 = 1 - \frac{SSE}{TSS}$$

Where, the residual sum of squares: represents the sum of the variance not explained by the model.

Total sum of squares: indicates the total variation in the dependent variable.

b1: Intercept, which represents the expected value of sales of conventional energy vehicles when sales of new energy electric vehicles are zero.

b2: Primary term coefficient indicating the impact of changes in sales of new energy electric vehicles on sales of conventional energy vehicles.

b3: Quadratic term coefficient indicating the effect of the change in the square of the sales of new energy electric vehicles on the sales of conventional energy vehicles.

form 5 estimated coefficient

	Estimate	SE	tStat	pValue
b1	5602.3	4.5459e-08	1.2324e+11	1.1845e-148
b2	27.098	1.2263e-05	2.2098e+06	3.3348e-82
b3	-0.070406	0.002687	-26.202	2.69e-13
b4	6.6904e-05	5.2837e-06	12.662	4.6798e-09

b5	-2.1086e-08	2.4683e-09	-8.5428	6.3262e-07

5.3.3 Conclusion

The specific data obtained were:
Number of observations: 17, error degrees of freedom: 14
Root mean square error: 377
R^2: 0.878, R^2 adjusted 0.861
F-statistic (constant model): 50.5, p-value = 3.99e-07

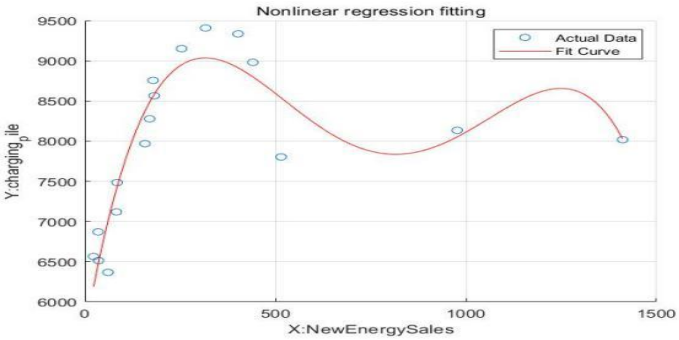

figure 7 Relationship between sales of new energy and sales of traditional energy vehicles

The multivariate nonlinear polynomials for this fit:

$$Y=5602.3+27.098X-0.70406X^2+6.6904e-05X^3-2.1086e-08*X^4$$

By interpreting the coefficients, the trend of the impact of the sales volume of new energy electric vehicles on the sales volume of conventional energy vehicles is analysed as follows:
$\beta_1 > 0$, indicating that the increase in sales of new energy electric vehicles is positively related to the increase in sales of traditional energy vehicles.
$\beta_1 < 0$, indicating that the increase in sales of new energy electric vehicles is slowing down from the increase in sales of conventional energy vehicles.

5.4 Problem 4 modelling and solution

5.4.1 Modelling requirements

1) Model assumptions
It is assumed that there is a negative correlation between price and quantity, i.e., when price rises, quantity demand falls.
It is assumed that there is a positive correlation between price and quantity, i.e. as price rises, quantity supplied rises.
2) Disincentive policies
Other countries' policies designated to target resistance to China's new energy electric vehicle development include:
① Trade Restrictions: Implement trade restrictions, such as raising import tariffs and imposing anti-dumping duties, in order to curb the competition of Chinese new energy

electric vehicles in the country's market.
(ii) Technical barriers: Setting technical standards or certification requirements, requiring new energy electric vehicles to meet higher technical standards in order to increase their entry barriers.
(iii) Negative publicity: Create a negative impression of China's new energy electric vehicles through the media or other means of publicity, thereby lowering their image in the minds of consumers.
④ Government Procurement Preferences: Reduce procurement of Chinese products by setting government procurement policies that favour the purchase of new energy electric vehicles produced in their own or other countries.
⑤ Export restrictions: Formulate export restriction policies to limit the export of China's new energy electric vehicles to this country, thus reducing its share of the international market.
(vi) Joint venture restrictions: Restrictions on joint ventures between Chinese companies and national companies, restricting Chinese new energy vehicle companies from establishing production bases in the country or participating in the development of national companies.

3) Modelling

1. Demand Modelling

Demand for electric vehicles is negatively correlated with price, i.e., as price rises, demand falls.
The negative publicity impact of the policy leads to a negative demand shock. $Q_d=a-bP+cG_{propaganda}$

2. Supply Modelling

The supply of electric vehicles is positively correlated with price, i.e., as price rises, supply increases.
Technical barriers to policy lead to a negative supply shock. $Q_s=d+eP-fG_{technology}$

3. Market balance

Market equilibrium occurs at the point where demand equals supply. $Q_d = Q_s$

4. Decision analysis and game theory model: $Q=f(P, P_{subsidy}, P_{tariff}, P_{standard})$

5.4.2 Supply and demand modelling solutions

1) Setting parameter values

Set specific values for model parameters a, b, c, d, e, f. Set the initial price P0.

2) Market equilibrium equations

Qd = Qs; substitute the demand and supply equations to solve for the price P and the corresponding equilibrium quantity in the market.

3) Categorical variable setting

Set two categorical variables:
X1: Presence of boycotts (yes/no)
X2: Did it lead to a slowdown in growth (yes/no)

4) Data collation

According to the implementation time of the boycott policy, the data are divided into

two periods before and after the boycott. Calculate the growth rate of the number of new energy vehicle exports and set the X2 variable according to whether the growth rate slows down or not.

5) Analysis of policy variables

Change $G_{propaganda}$ and $G_{technology}$ to observe changes in price, demand and supply. Analyse the impact of policy variables on the market.

6) Sensitivity analysis

Varying the values of the model parameters and observing the sensitivity of the model results, with a special focus on the impact of policy parameters on the market equilibrium.

7) Kruskal-Wallis H test

Kruskal-Wallis H-test for X1 and X2, to test whether X1 has a significant effect on the number of new energy vehicle exports, and whether X2 has a significant effect on whether the growth rate slows down.

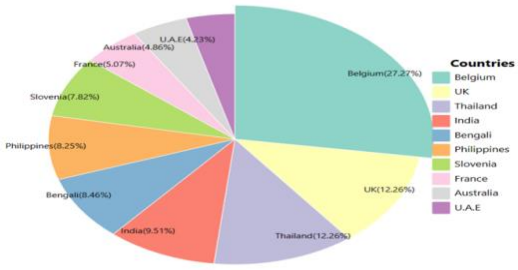

figure 8 China's exports of EVs

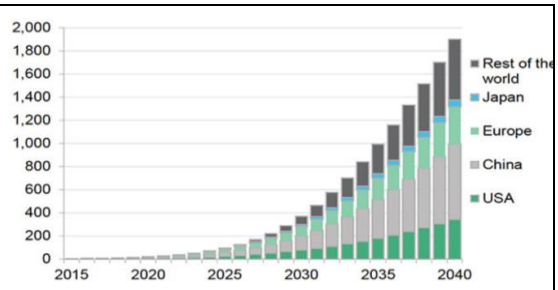

figure 9 Global EVs demands (TWh)

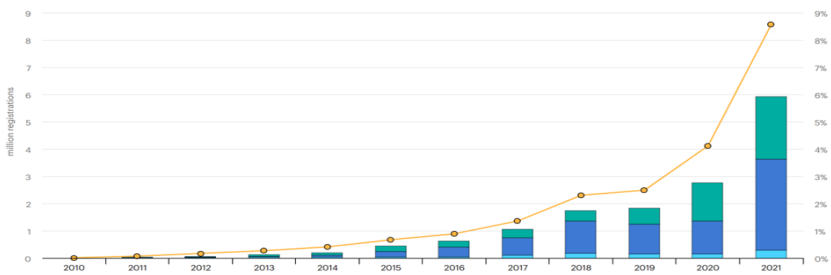

figure 10 Global sales and sales market share of electric cars, 2010-2021

5.4.3 Decision analysis and game theory modelling

1) Participants

① Chinese government ② international partners (other governments, international organisations, etc.) ③ Chinese new energy electric vehicle enterprises.

2) Decision-making variables

①Government policy decisions (subsidy levels, tariff levels, environmental standards)

②Enterprise market strategy (export volume, domestic market share)

3) The gaming process

The Chinese government and international partners negotiate and formulate international policies through a gaming process. Firms develop their market strategies under international policies.

4) Utility function

GoC: Consider the overall utility of various factors such as economy, environmental protection and industrial competition.
International partners: Consider the overall utility of factors such as cooperation, trade balances and technological innovation.
Firms: the overall utility of considering multiple factors such as market share, profits, and international reputation.
5) Evaluation of game solutions
Analyse Nash equilibrium to understand the optimal combination of decisions for different participants. Consider the balance between co-operation and competition and assess the benefits to each party.

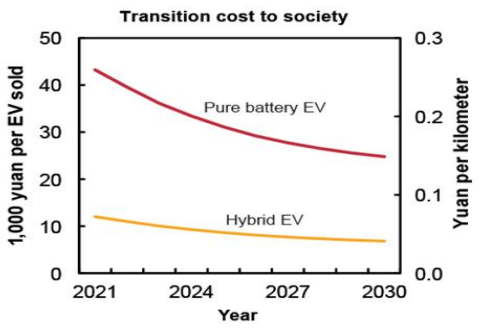

figure 11 Transition cost to society

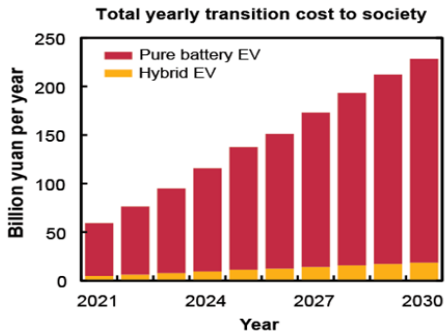

figure 12 Total yearly transition cost to society

5.4.4 Conclusion

1. Limited market access: Trade restrictions and technical barriers may make it more difficult for Chinese new energy electric vehicles to enter the markets of affected countries. This could lead to sales blockages and slow down the development of Chinese companies in that country's market.
2. Decreased price competitiveness: Subsidy restrictions could lead to a decrease in the price competitiveness of Chinese new energy electric vehicles in the markets of the affected countries. This could affect consumers' vehicle purchase decisions, especially in price-sensitive markets.
3. Damage to image: Negative publicity impacts negatively on the brand image of China's new energy electric vehicles. This may lead to a decrease in consumer trust in Chinese brands and affect their market share in the international market.
4. Strategic adjustments: In the face of these policies, Chinese new energy EV companies need to adjust their strategies, look for other market opportunities, strengthen the development of the domestic market, or strengthen cooperation with other countries.
5. Industrial restructuring: These policies will lead to the adjustment of China's new energy electric vehicle industry structure, companies will look for new markets, technology partners, or adjust the product positioning and innovation direction.
Overall, these policies may bring a series of challenges to China's new energy EV industry, but they may also prompt companies to focus more on technological innovation, internationalisation strategies and market diversification to adapt to the

changing international environment. Chinese new energy EV companies need to be flexible and adjust their strategies in the face of these challenges to ensure sustainable development.

5.5 Problem 5 modelling and solution

5.5.1 Modelling system dynamics

1) Model assumptions

1. Different countries have different standards for pollutant emissions, and it is assumed that the vehicle emission standards in the city are the Chinese national standards.
2. Assume that the city of one million people is Xiamen, Fujian Province, China, with a population of 5 million and a total of 2 million vehicles.
3. Assume that 190,000 motor vehicles are licensed in Xiamen in this year. Of these, 130,000 are passenger cars and 3,300 are new energy vehicles.

2) Modelling

System dynamics modelling [15] focuses on the dynamic process of system evolution over time. In this scenario, we simulate the changes over time in variables such as the number of vehicles, traffic flow, emissions, air quality, noise level and energy mix. Life Cycle Assessment (LCA) is a comprehensive methodology for assessing the potential environmental impacts of a product or service throughout its life cycle. Consideration in this topic includes the manufacturing, use and disposal stages. Such comprehensiveness helps to capture the multifaceted environmental impacts of vehicles, not just the emissions during the use phase. See appendix for data sources.

3) Extended System Dynamics Model

dynamic equation (math.)

Number of vehicles:

$$\frac{dV}{dt} = Sales_{Gasoline} + Sales_{NEVs} - Scrap_{Gasoline} + Scrap_{NEVs}$$

Traffic flow:

$$\frac{dT}{dt} = F(V, \text{Other transportation factors})$$

Conventional Fuel Vehicles and NEVs Emissions:

$$E_{Gasoline} = V_{Gasoline} * Emission_{Gasoline}$$

$$E_{NEVs} = V_{NEVs} * Emission_{NEVs}$$

Air Quality:

$$\frac{dAQ}{dt} = g(E_{Gasoline}, E_{NEVs}, \text{Other pollution sources})$$

Noise level:

$$\frac{dNL}{dt} = h(V, \text{Other noise factors})$$

Energy mix:

$$\frac{dES}{dt} = i(V, \text{Other energy sources factor})$$

Manufacturing, use and disposal phase costs:

$$M_{Gasoline} = j(V_{Gasoline}, \text{Other manufacturing factors})$$

$$M_{NEVs} = k(V_{NEVs}, \text{Other manufacturing factors})$$

$$M_{Gasoline} = iV_{Gasoline}, \text{Other abandonment factors})$$

$$M_{NEVs} = M(V_{NEVs}, , \text{Other abandonment factors})$$

4) Life cycle stages

1. Manufacturing phase:

Consider the manufacturing process for components such as batteries, bodies, motors, etc. Consider the environmental impact of raw material extraction, manufacturing processes, transport, etc.

2. Stage of use:

Consider the energy consumption, emissions of NEVs during their use. Consider the impact of battery charging and the construction of charging facilities on the grid.

3. Abandonment stage:

Consider disposal methods during the discard phase, including recycling and disposal of batteries. Consider the potential environmental impacts of the disposal process.

5.5.3 Solving with Life Cycle Assessment (LCA)

1) Manufacturing phase

Functional unit definition: Manufacture of a new energy vehicle

Specific considerations:

Body: length between 3800mm and 4300mm, width between 1600mm and 1800mm, height between 1400mm and 1600mm

Battery capacity: between 10 kWh and 100 kWh

Motor power: 150KW

2) Use phase

Functional unit definition: the number of miles travelled by each vehicle over its entire useful life.

Specific considerations:

National standard voltage of charging pile interface: output voltage range within 300---800V

AC Charging Post (220V, 380V): 7KW/3.5KW

DC charging post: 60kW,450V ,800V ,1000V

Small power DC pile: 7KW-30KW

Tesla V3: liquid-cooled gunline, 250kW = 400V * 625A.

The average life of a vehicle is 15 years with 600,000 kilometres driven

While conventional fuel vehicles produce 2.36 kilograms of carbon dioxide per litre of petrol, new energy electric vehicles produce 0.6 kilograms of carbon dioxide per kilowatt-hour of electricity.

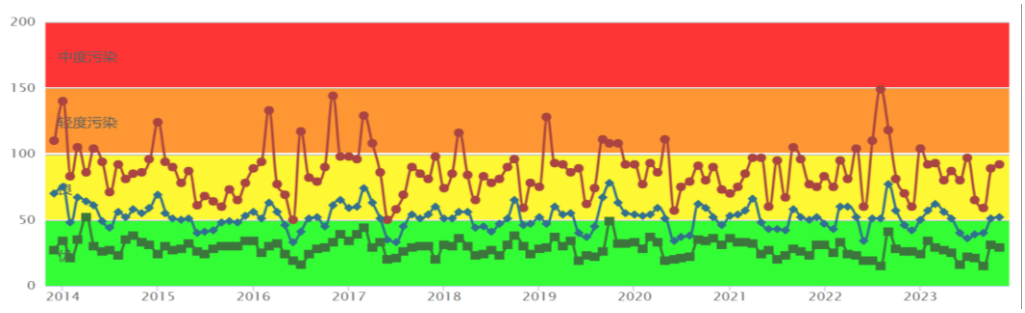

figure 13 Monthly trend of Xiamen Air Quality Index (AQI) level changes

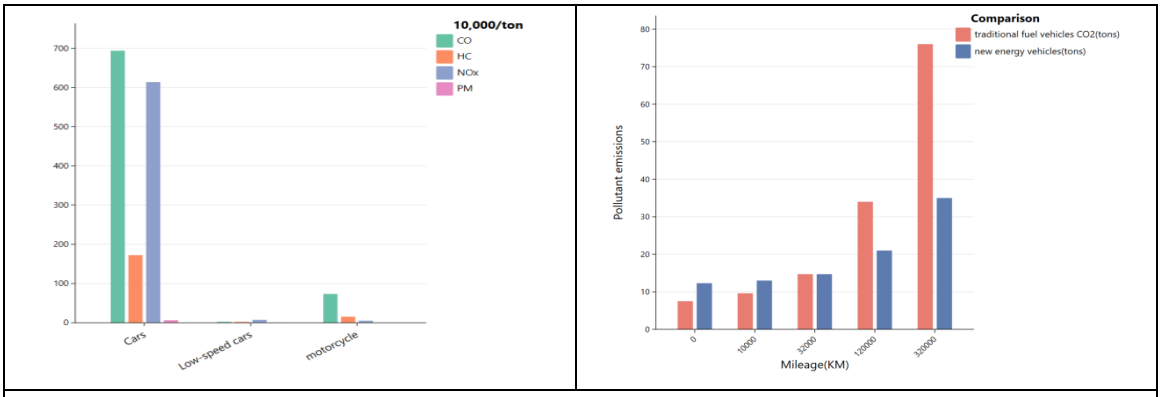

figure 14 The relationship between emissions and mileage

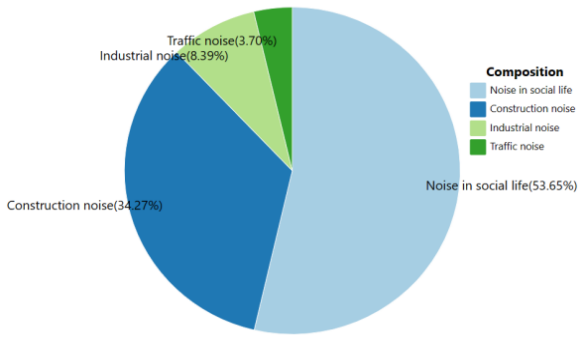

figure 15 Urban noise composition

form 6 Urban noise standards

Noise limits (Unit: dB(A))						
Ribbon Categories.	Class 0	Class 1	Class 2	Class 3	Class 4a	Class 4b
Days	≤ 50	≤ 55	≤ 60	≤ 65	≤ 70	≤ 70
Nigts	≤ 40	≤ 45	≤ 50	≤ 55	≤ 55	≤ 60

3) Abandonment stage

Functional unit definition: Dispose of a discarded new energy vehicle.

Specific considerations:

In 2021, the recycling volume of power battery was at the level of 200,000 tonnes, and in 2022, the recycling volume of power battery was at the level of 400,000 tonnes. Lithium batteries (ternary batteries, lithium iron phosphate batteries, lithium cobalt acid

batteries) retired 579,000 tonnes, it is expected that the amount of retired batteries in 2030 will reach 437GWh, can be recycled lithium carbonate, iron phosphate, nickel sulfate, cobalt sulfate were 230,000,000, 390,000,000, 3.46 million, 270,000 tonnes, recycling market value of 169.5 billion yuan.

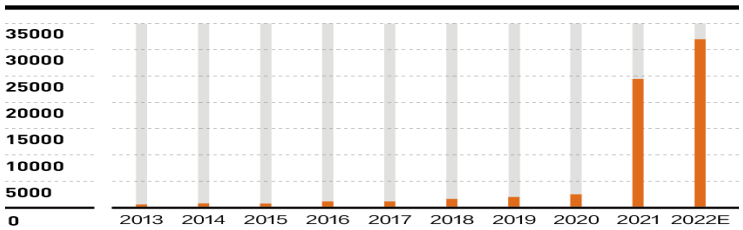

figure 16 Trends in Recycling Enterprise Registration from 2013 to 2022

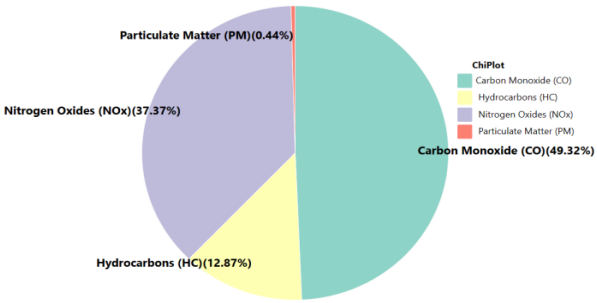

figure 17 The emission structure and amount of traditional fuel vehicles

5.5.3 Conclusion

Air Quality Impacts:

Positive Impact:

1. Reduced Tailpipe Emissions: New Energy Vehicles (NEVs), primarily electric vehicles, exhibit no tailpipe emissions, enhancing environmental friendliness compared to traditional fuel vehicles.
2. Improved Air Quality: Mitigation of tailpipe emissions contributes positively to reducing air pollutant levels in cities.

Negative Impact:

1. Battery Manufacturing and Disposal Environmental Impacts: Processes involved in battery manufacturing and disposal may introduce environmental concerns, including resource extraction, energy consumption, and disposal.
2. Pollution from Electricity Production: The positive air quality impact of NEVs may be compromised if urban electricity relies heavily on highly polluting sources.

Noise Pollution Impact:

Positive Impact:

Reduced Traffic Noise: Electric vehicles, particularly at low speeds, contribute to lowered traffic noise levels, enhancing urban residents' quality of life.

Negative Impact:

Safety Concerns from Quiet Operation: The diminished noise of electric vehicles at low speeds poses potential safety hazards for pedestrians and other road users.

Urban Planning Impacts:

Positive Impact:

Promoting Green Mobility: NEV promotion facilitates cities in achieving green mobility objectives by incentivizing environmentally friendly transport facilities like electric vehicle charging stations and bike paths.

Negative Impact:

Infrastructure Development Challenges: The promotion of NEVs necessitates corresponding charging facilities and infrastructure support, potentially introducing challenges to urban planning and construction.

Energy Mix Impacts:

Positive Impact:

Facilitating Renewable Energy Adoption: NEV adoption may drive cities to expedite the transition to renewable energy sources, aligning with the energy requirements of electric vehicles and mitigating environmental impacts.

Negative Impact:

Environmental Consequences of Energy-Intensive Electricity: The push for electric vehicles may impose additional environmental burdens if urban electricity predominantly originates from energy-intensive and polluting sources.

5.6 Question 6 - Open Letter

Dear Fellow Citizens:

Salutations,

In the face of escalating global environmental challenges and the urgency to address climate change, I extend warm greetings to you. Today, I aim to elucidate the myriad advantages of new energy electric vehicles and their profound implications for our urban landscapes and the global community.

Against the backdrop of intensifying climate concerns, it is imperative to explore more environmentally sustainable modes of transportation. New energy electric vehicles emerge as an optimal choice in this context. Distinct from conventional fuel-driven counterparts, electric vehicles utilize electricity as their power source, devoid of emissions such as exhaust fumes and particulate matter. This absence of pollutants not only augments our health but also has positive ramifications for the broader ecosystem. Our research substantiates that the proliferation of electric vehicles correlates with a substantial enhancement in urban air quality and a noteworthy reduction in carbon emissions. Moreover, these vehicles exhibit minimal noise pollution, contributing to a quieter and more habitable urban environment. Beyond environmental considerations, the adoption of electric vehicles significantly diminishes reliance on petroleum resources, thereby safeguarding national energy security and making a constructive

impact on global climate change.

New energy electric vehicles not only epitomize a more environmentally conscientious and sustainable mode of transportation but also signify an inexorable future trend. With advancing technology poised to become more ubiquitous, electric vehicles are poised to emerge as a principal mode of transportation. Opting for electric vehicles not only affords us a cleaner and quieter driving experience but also aligns us with a global commitment to ecological well-being.

The electric vehicle industry assumes a pivotal role on the international stage. Aligned with the prevailing global emphasis on environmental preservation and sustainable development, the electric vehicle sector experiences rapid expansion, providing abundant employment opportunities and economic dividends for nations worldwide. Concurrently, the growth of the electric vehicle industry catalyzes advancements in associated sectors such as electric power, batteries, and materials, thereby stimulating global economic development. Moreover, the evolution of the electric vehicle industry propels scientific and technological innovation, fostering industrial upgrades. The continual advancements and widespread adoption of electric vehicle technology herald not only technological prowess but also lay the foundation for future national development.

In conclusion, I extend a heartfelt invitation to each citizen to partake in this green revolution. Let us collectively choose electric vehicles, contributing significantly to the betterment of our city and our planet. Our individual choices, though seemingly small, possess the transformative power to create a cleaner and more sustainable world for generations to come.

Thank you for your attention and unwavering support.

VI. Model evaluation

6.1 Model benefits

1. Multiple linear regression: the complexity of the model can be controlled by adjusting the weights of the independent variables.
2. Grey prediction model: It is more suitable for emerging industries such as new energy vehicles because it is suitable for small samples and high uncertainty situations. It is more suitable for data that lacks regularity, is unstable or difficult to capture.
3. Multivariate non-linear regression: algorithms are easy to implement and deploy,

perform efficiently and accurately.
4. System dynamics model [16]: it can better reflect the dynamic process of the growth of new energy vehicles and the interaction between various factors.

6.2 Modelling shortcomings

1. Multiple linear regression: for cases where there is covariance between the independent variables, the results of the model may be unstable and the interpretation of the variables may be inaccurate.
2. Grey forecasting model: only suitable for short to medium term forecasting, only suitable for approximating exponential growth. It is necessary to construct a first-order ordinary differential equation to solve the functional expression of the fitted function.
3. Multiple non-linear regression: discrete independent variable data need to be used by generating dummy variables
4. The construction of the prediction model is relatively single, and the analysis of the influencing factors of new energy vehicle sales is not comprehensive enough. Due to the short development time of new energy vehicles, the data collection is small and incomplete, and the accuracy of the prediction results needs further in-depth research and improvement.

VII. Model roll-out

The model developed in this paper . After being put into use in real life, together with the comprehensive consideration of other factors in reality, it can analyse the prediction of other factors on the development of new energy electric vehicles and provide reference for decision makers. In the subsequent research, the sample size should be increased, and the sample should be classified and stratified, so that the results of the prediction of the development of new energy electric vehicles will be more accurate.

VIII. References

[1] Yang Z. Deepening global co-operation to promote sustainable industrial development[N]. China Energy News,2023-09-18(012).DOI10.28693n.cnki.nshca.2023.001581.
[2] ZHAO Yonggang,YU Zhen. Research on the evaluation of factors influencing satisfaction of different car brands based on entropy weight-variance method[J]. China Automobile,2021(10)4-9.
[3] Yu Yan,Yuan Xinyi. China's new energy vehicle market sales forecast[J]. Internal Combustion Engines and Accessories,2021(06):179-180.DOI:10.19475/j.cnki.issn1674-957x.2021.06.086.
[4] Wang Lanlan. Application of improved GM(1,1) model in new energy vehicle sales forecast[J]. Journal of Economic Research,2023(15):46-50.
[5] Zhang Shuangni. Forecast of new energy vehicle market development trend based on multiple regression model[J]. Decision Exploration (in Chinese), 2019(01):77.

[6] QI Hai-Ping,SHIN Xi-Ping.Implementation of Kruskal-Wallis H test for mean rank multiple comparisons in SPSS software[J]. Journal of Lanzhou Institute of Technology,2015,22(02):76-78.

[7] ZHAO Lei, ZHANG Yannan, DING Jun et al. Analysis of new energy vehicle industry policy differences under the perspective of supply and demand[J]. Journal of China University of Petroleum (Social Science Edition),2022,38(01):9-16.DOI:10.13216/j.cnki.upcjess.2022.01.0002.

[8] Wang Lei. Relationship between green low-carbon recycling economy, double integral policy and new energy vehicle industry - A simulation study based on system dynamics model[J]. Journal of Tongling Institute of Vocational Technology,2021,20(04):28-32+61.DOI:10.16789/j.cnki.1671-752x.2021.04.00.

[9] Song Xiaocong, Deng Chenning, Shen Peng et al. Environmental impact and carbon footprint analysis of pure electric vehicles based on life cycle assessment[J]. Environmental Science Research,2023,36(11)2179-2188.DOI10.13198j.issn.1001-6929.2023.08.15.

[10] Shen Kasa, Ma Heping. Semi-hidden Runge-Kutta method for ordinary differential equations[J]. Journal of Applied and Computational Mathematics,2017,31(03)303-315.

[11] Yuan Xueju. Research on energy transition and high-quality development strategy based on carbon neutral target - taking Shaanxi Province as an example [C] Shanxi Province CUEC Research Institute. Proceedings of the Symposium on Rural Revitalisation and Education Development - Research on Industrial Revitalisation. [Publisher unknown], 20233.DOI10.26914c.cnkihy.2023.047503.

[12] Jingtao Chen, Mang Xu. Simulation of the development trend of new energy electric vehicles in Beijing[C]// China Association for Science and Technology,Ministry of Transport,Chinese Academy of Engineering,Hubei Provincial People's Government.Proceedings of the World Conference on Transport 2023 (WTC2023)(Lower Volume). People's Transportation Press Co. Ltd,2023:7.DOI:10.26914/c.cnkihy.2023.019459.

[13] Wang Shuangwen. Empirical analysis of logistics influencing factors based on multiple linear regression model--Taking Shandong Province as an example[J]. China Collective Economy,2023(31)89-92.

[14] ZHU Manyu,ZHANG Zheng. Research on new energy vehicle sales prediction based on GM(1,1) model[J]. Logistics Engineering and Management,2023,45(02):138-141.

[15] Zhang Yu. Research on the optimisation of new energy vehicle industry policy in Beijing [D]. North China Electric Power University (Beijing), 2023.DOI:10.27140/d.cnki.ghbbu.2022.000481.

[16] Jingtao Chen, Mang Xu. Simulation of the development trend of new energy electric vehicles in Beijing[C]// China Association for Science and Technology,Ministry of Transport,Chinese Academy of Engineering,Hubei Provincial People's Government.Proceedings of the World Conference on Transport 2023 (WTC2023) (Lower volume). People's Transportation Press Co. Ltd,2023:7.DOI:10.26914/c.cnkihy.2023.019459.